

Why Benders finds the schedule but struggles to prove it

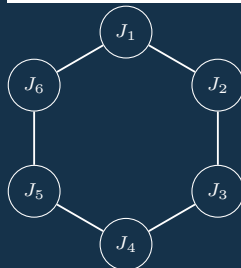
Grouping gaps, coverage bounds, and exact certification
in the Job Sequencing and Tool Switching Problem

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order the jobs
keep useful tools resident

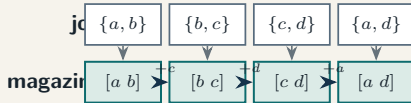
A small magazine turns production order into a cost

MODEL ■ empty start shown

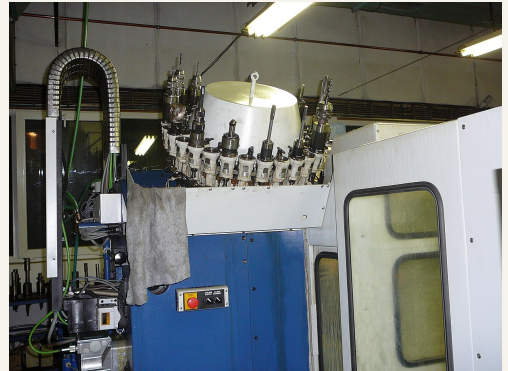
Jobs $j \in J$ require tool sets $T_j \subseteq T$; the magazine holds at most b tools. For a job order π and magazine states M_i ,

$$T_{\pi(i)} \subseteq M_i, \quad |M_i| \leq b, \quad M_0 = \emptyset,$$

$$\text{cost}(\pi, M) = \sum_{i=1}^n |M_i \setminus M_{i-1}|.$$



The schedule pays when a needed tool is absent; a poor order makes tools leave and return.



A CNC tool magazine: limited slots, many candidate tools.

Loading is exact once the order is fixed

INNER PROBLEM ▪ free-initial value



The asymmetry

fixed-order loading

polynomial

choose the job order

NP-hard
for every
fixed $b \geq 2$

When a required tool is missing, insert it; if the magazine is full, evict the resident tool whose next use is *furthest* away.

KTNS(π) in $O(n|\mathcal{U}|)$

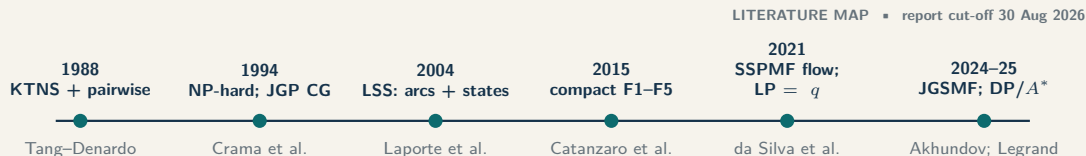
oracle
exact

ordering
combinatorial



This separation motivates both the practical grouping shortcut and an exact Benders decomposition.

Exact SSP methods are a search for a useful lower bound



What the compact line learned

- ▶ LSS repaired a zero LP with arc, magazine-state and switching variables.
- ▶ F5 is at least as strong as F4 and is the 2015 recommendation.
- ▶ SSPMF routes each tool through positions; its relaxation is the coverage count q when every declared tool is used.

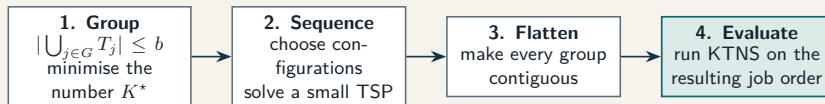
Laporte et al. (2004); Catanzaro et al. (2015); da Silva et al. (2021/2024); Akhundov–Ostrowski (2024); Legrand et al. (2025).
JGP—declared tool is used

What this campaign does—and does not—compare

- ▶ Reimplemented baselines: LSS, F4 and SSPMF.
- ▶ Not implemented: F5, strengthened JGSMF, or the newer DP/ A^* frontier.
- ▶ Therefore this is a diagnosis of tested formulations, *not* a SOTA ranking.

The practical shortcut is group, sequence, then reload optimally

GROUPING LITERATURE • optimistic envelope



Why it is attractive

- ▶ JGP has exact set-covering / representative formulations.
- ▶ It is effective in industrial colour-printing studies.
- ▶ The classical worst-case guarantee is multiplicative: at most b times optimum.

Quantity audited here

$$H = \min_{\substack{\pi \text{ induced by a} \\ K^* \text{-grouping}}} \text{KTNS}(\pi).$$

This minimises over every grouping, group order, and within-group order: bad tie-breaking has been removed.

If this best possible envelope is suboptimal, the loss comes from minimum-group contiguity itself.

Coverage separates first appearances from returns

THEORY ▪ free initial magazine

$$q = |\mathcal{U}| - b$$

unavoidable first insertions after the free fill

tight



loose



evict, then reinsert

Every schedule decomposes into

$$Z^* = q + R_{\text{opt}}, \quad H = q + R_{\text{grouped}},$$

where R counts returns after a tool's first appearance.**coverage-tight** $Z^* = q$: each tool can occupy one residency block**coverage-loose** $Z^* > q$: some tool must occupy at least two blocks

The gap is exactly the excess reinsertion forced by grouping:

$$H - Z^* = R_{\text{grouped}} - R_{\text{opt}}.$$

Coverage bound and residence-block identity: report, Sections 2.6 and 4.2. Empty-start values add the common initial load and leave the gap unchanged.

Four jobs force one avoidable reinsertion

SEED GADGET ▪ $b = 4, |\mathcal{U}| = 7$

$A = \{1, 2\}, \quad B = \{3, 4\},$
 $C = \{1, 3, 5, 6\}, \quad D = \{2, 5, 6, 7\}.$

Only A and B can share a magazine:

$K^* = 3, \quad \{A, B\}, \{C\}, \{D\}.$

Hence every minimum-group order makes A, B consecutive.



Unrestricted optimum: interleave the pair



$q = 7 - 4 = 3, \quad Z^* = 3.$

Grouped orders: six possibilities up to reversal

order	$ABCD$	$ABDC$	$BACD$	$BADC$	$CABD$	$CBAD$
KTNS	4	5	4	5	4	4

$Z^* = 3$ but $H = 4$: the minimum-group restriction destroys the zero-reinsertion interleaving.

Report, Proposition 4.11 (smallest sub-optimal instance); all six values are independently reproduced by the theorem verifier.

Connect g seeds without permitting cross-copy groups

For each private copy i , add a universal tool u and a private marker v_i :

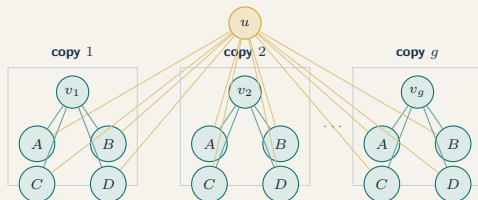
$$A_i = \{u, v_i, 1_i, 2_i\}, \quad B_i = \{u, v_i, 3_i, 4_i\},$$

$$C_i = \{u, v_i, 1_i, 3_i, 5_i, 6_i\}, \quad D_i = \{u, v_i, 2_i, 5_i, 6_i, 7_i\}.$$

$$|\mathcal{U}| = 7g + g + 1 = 8g + 1.$$

Roles. u connects every job; v_i reserves one slot and identifies copy i .

CONNECTED FAMILY $\widetilde{S}_g$ ■ $b = 6$



Every job contains u : both the job-intersection graph and the bipartite incidence graph are connected.

The unique minimum grouping has three groups per copy

PROOF 1/3 ▪ $K^* = 3g$
 $A_i \cup B_i$

u	v_i	1_i	2_i	3_i	4_i
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1. A_i, B_i together use exactly six tools, so the pair is feasible.

 $A_i \cup A_j$

u	v_i	1_i	2_i	v_j	1_j	2_j
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2. Small jobs from different copies need seven tools, so they cannot mix.

 $C_i \text{ or } D_i$

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3. Each C_i, D_i fills the magazine, and every other distinct job adds a missing tool; both are singleton groups.

$$\{A_i, B_i\}, \{C_i\}, \{D_i\} \quad (i = 1, \dots, g)$$

The grouping is unique, so no alternative optimum grouping can repair the forced adjacency.

An unrestricted order attains coverage exactly

PROOF 2/3 ▪ $Z^* = 8g - 5$

Coverage gives

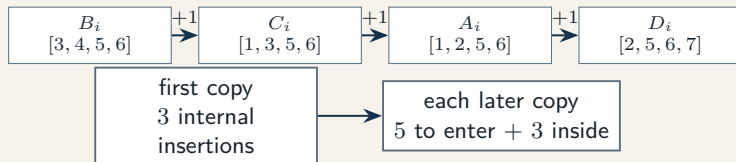
$$Z^* \geq q = (8g + 1) - 6 = 8g - 5.$$

Process copies consecutively, each in the seed order

$$B_i, C_i, A_i, D_i.$$

Keep u throughout; within copy i , keep v_i throughout.

One copy, suppressing the fixed slots u, v_i



$$3 + 8(g - 1) = 8g - 5.$$

No private tool is reinserted: the construction meets the coverage lower bound, so $Z^* = q$.

Minimum grouping forces one return in every copy

PROOF 3/3 ▪ $H = 9g - 5$

Every admitted order has A_i, B_i consecutive. Projecting any loading onto copy i leaves only four original-tool slots, so the seed lemma forces one private reinsertion.

Let x_i count copy- i original tools in the free initial magazine; let y count initially loaded tools among $\{u, v_1, \dots, v_g\}$. Then

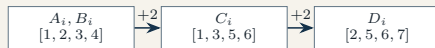
$$H \geq \sum_{i=1}^g (8 - x_i) + (g + 1 - y),$$

$$\sum_i x_i + y \leq 6,$$

$$\implies H \geq 9g - 5.$$

The grouped order A_i, B_i, C_i, D_i , copy by copy, attains the bound:

$$4 + 9(g - 1) = 9g - 5.$$

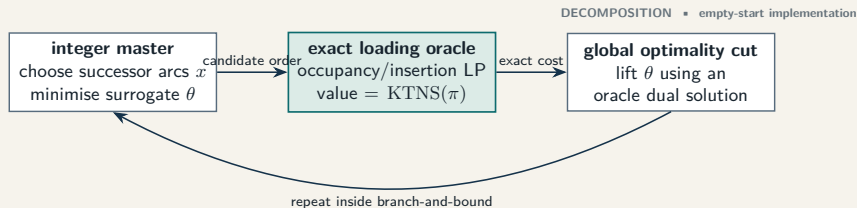


$$K^* = 3g, \quad Z^* = 8g - 5, \quad H = 9g - 5, \quad H - Z^* = g.$$

Connected, fixed capacity 6, unbounded additive gap; $H/Z^* \rightarrow 9/8$.

The lost object is a long-range residency decision: minimum grouping forces a return that the unrestricted order avoids.

Benders asks whether an exact oracle yields a useful certificate



Theory convention



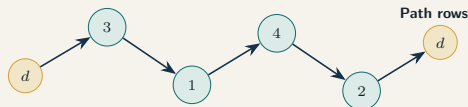
Master convention

$$q = |\mathcal{U}| - b, \quad Z^* = q \text{ means tight.}$$

$$\theta \geq |\mathcal{U}|, \quad Z_{\emptyset}^* = |\mathcal{U}| \text{ means tight.}$$

Prediction: coverage certifies tight cases immediately; loose cases require the master to certify a reinsertion.

The master is a Hamiltonian-path skeleton with one cost surrogate



$x_{ij} = 1$ iff j immediately follows i
depot d marks start and end

BBC MASTER ■ exact rows shown

Path rows

$$\sum_{i \neq j} x_{ij} = 1, \quad \sum_{j \neq i} x_{ij} = 1, \quad \sum_{i,j \in S} x_{ij} \leq |S| - 1.$$

Always-on lower bounds

$$\theta \geq \sum_{i,j \in J} \underbrace{\max(0, |T_i \cup T_j| - b)}_{w_{ij}} x_{ij} + \sum_{j \in J} |T_j| x_{dj}, \quad \theta \geq |\mathcal{U}|.$$

$$\min \theta, \quad x_{ij} \in \{0, 1\}, \quad \theta \geq 0.$$

Optional triplets

$$\theta \geq w_{ijk}(x_{ij} + x_{jk} - 1), \quad w_{ijk} = \max(0, |T_i \cup T_j \cup T_k| - b).$$

Implementation scope: subtour constraints are separated lazily at integer candidates; fractional SEC separation was not tested.

The loading LP is integral; its dual becomes an arc cut

BBC ORACLE ■ fixed candidate $\bar{x}$

y_{jt} says tool t is present at job j ; z_{jt} counts its insertion.

For any dual-feasible $(\lambda, \lambda^d, \mu, \nu)$,

$$\begin{aligned}
 \min \quad & \sum_{j,t} z_{jt} \\
 \text{s.t.} \quad & y_{jt} \geq 1 && t \in T_j, \\
 & \sum_t y_{jt} \leq b && \forall j, \\
 & z_{jt} \geq y_{jt} - y_{it} - (1 - \bar{x}_{ij}) && i \neq j, \\
 & z_{jt} \geq y_{jt} - (1 - \bar{x}_{dj}) && \forall j, t, \\
 & y, z \geq 0.
 \end{aligned}$$

For a fixed path the consecutive-ones structure makes this relaxation integral; its value equals KTNS.

$$\begin{aligned}
 \theta \geq \sum_{i \neq j, t} \lambda_{ijt} (x_{ij} - 1) + \sum_{j, t} \lambda_{jt}^d (x_{dj} - 1) \\
 - b \sum_j \mu_j + \sum_j \sum_{t \in T_j} \nu_{jt}.
 \end{aligned}$$

- ▶ Valid for every order; tight at $\bar{x}$ for an optimal dual.
- ▶ Coefficient of x_{ij} is $\sum_t \lambda_{ijt}$: every learned row is arc-supported.
- ▶ The depot term is essential. Its earlier omission generated 193 random counterexamples; it was restored before the campaign.

The oracle is exact. The open issue is expressive power: can adjacency-indexed cuts price a cost created across a span?

Nine BBC regimes change where the same information enters

ABLATION DESIGN ■ fixed denominator 1,410

F = fractional LP-dual separation; T = triplets; C = conflict row; H = HGS start + seed cut; P = core-point/Pareto cut.

configuration	exact implementation change	proved / 1,410
BBC-LP	LP-dual optimality cut at integer candidates	724
BBC-K	KTNS combinatorial path cut at integer candidates	722
BBC-LP+T	LP cut plus $O(n^3)$ triplet rows	723
BBC-LP+F	LP-dual user cuts at fractional nodes	655
BBC-K+F	combinatorial integer cut plus fractional LP cuts	657
BBC-LP+F+C	conflict-graph constant root row	654
BBC-LP+F+H	two-second HGS start plus one seeded Benders cut	686
BBC-LP+F+P	ordinary and Papadakos/core-point cuts	618
BBC-LP+ACC	$F + C + H + P$ together	652

Interpretation guard. Enabling F also disables CPLEX presolve. H adds a seed cut and about two seconds of preprocessing before the recorded CPLEX timer. These are implementation regimes, not perfectly isolated causal treatments.

The primary ledger is complete; the comparison has fixed denominators

1,410
canonical instances

12
configurations

16,920
of 16,920 rows present

Primary protocol

- ▶ one hour and 16 GiB per run
- ▶ CPLEX 22.1.1 on one homogeneous LIMOS node pool
- ▶ time and memory limits remain unsolved outcomes
- ▶ separate 32-GiB LSS sensitivity excluded; zero additional optima

Thread reality

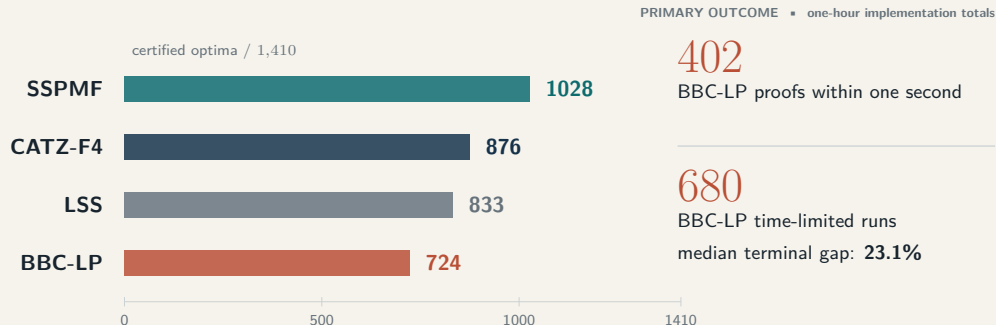
Four cores allocated. BBC used one CPLEX thread; compact baselines used four. Denominators match; runtimes do not.

CAMPAIGN PROTOCOL ■ canonical data only

Independent checks

- ▶ KTNS vs exact state DP: 349,872 sequence pairs, zero disagreement.
- ▶ 988 instances proved by at least two methods: 41,514 method pairs, zero cost disagreement.
- ▶ 1,107 instances have a campaign-proved optimum; 303 remain unknown.

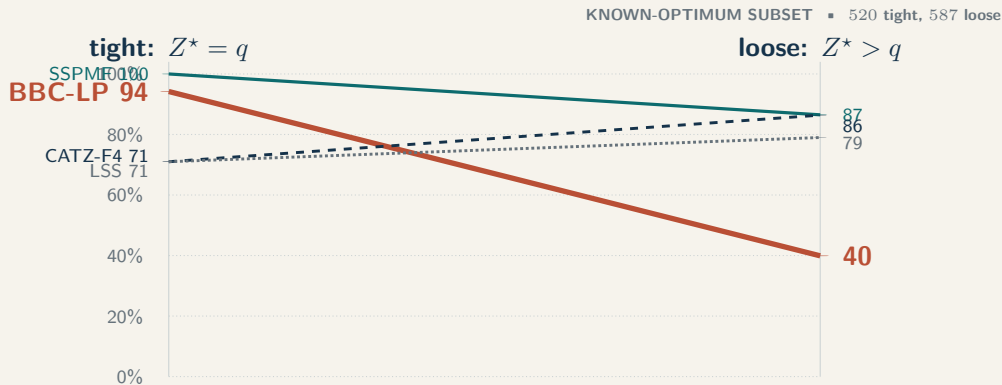
Under this protocol, the compact baselines close more



The method is not uniformly slow: it often finishes immediately, then has a long certificate tail. Because effective threads differ, the bars describe the tested implementations under the allocation—not intrinsic algorithm speed.

The next question is not merely “who wins?”: which structural cases does the Benders master fail to certify?

Closure is strongly associated with coverage tightness



BBC-LP: 94.2% tight versus 39.9% loose (**-54.4 points**); controls show association, not universal hardness.

The primal is often right while the certificate remains at coverage

CERTIFICATE DIAGNOSIS ■ three independent measurements

87.2%

9,067/10,392 recorded Benders roots equal coverage exactly.

58.2%

223/383 unfinished BBC-LP runs with a known optimum already hold it.

23.1%

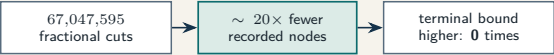
Median terminal gap across BBC-LP's 680 time-limited runs.

The incumbent can already equal the optimum while the lower bound remains near coverage.

Together these measurements support a certificate bottleneck in this implementation; they do not prove that search is never limiting.

The implemented fractional regime cut nodes, not terminal bounds

Implemented fractional regime F



On 1,404 paired terminal bounds:

	higher with F	equal	lower with F
instances	0	1,226	178

Certified totals move from 724 without F to 655 with the implemented F regime.

WITHIN-BBC ABLATION • one-thread fixed budget

Against BBC-LP+F = 655

addition	solved	change
H	686	+31
C	654	-1
P	618	-37
$C + H + P$	652	-3
remove F	724	+69

F also turns presolve off. H supplies both a start and a seed cut, with about two seconds outside the recorded CPLEX timer. The differences are not clean feature-only causal estimates.

The tested loading-dual augmentations did not improve the terminal certificate within budget; fewer nodes were not a stronger proof.

CONCLUSION

The evidence points to reinsertion across a span

Proved

Minimum grouping has a connected, fixed-capacity family with

$$H - Z^* = g.$$

The loss is one forced return per copy, not bad tie-breaking.

Measured

BBC-LP certifies 94.2% of tight cases and 39.9% of loose ones. Roots usually equal coverage; many unfinished incumbents are already optimal.

WHAT IS PROVED, MEASURED, AND NEXT

Next controlled test

1. match threads, presolve, and all preprocessing time;
2. add fractional subtour separation as a control;
3. test nonlocal, configuration-consistent columns or inequalities.

Success criterion: raise the root materially above coverage on many loose instances *and* turn that gain into more fixed-budget proofs.

An exact oracle finds the schedule; a useful formulation must also explain why the tool had to return.

Backup 1: the connected-gap proof as one ledger

Construction

$b = 6$; copy i has seven original tools, marker v_i , and universal u :

$$\begin{array}{l|l} A_i & u, v_i, 1_i, 2_i \\ B_i & u, v_i, 3_i, 4_i \\ C_i & u, v_i, 1_i, 3_i, 5_i, 6_i \\ D_i & u, v_i, 2_i, 5_i, 6_i, 7_i \end{array}$$

Only A_i, B_i share; C_i, D_i are singleton; cross-copy pairs overflow. Therefore $K^* = 3g$. Coverage and the order

B_i, C_i, A_i, D_i give

$$q = |\mathcal{U}| - b = 8g - 5, \quad Z^* = q.$$

USE FOR QUESTIONS ON K^* , Z^* , OR THE LOWER BOUND

Why every admitted order pays g more

Minimum grouping forces A_i, B_i consecutive. Let x_i be copy- i original tools in the initial magazine and y the initially loaded common/marker tools.

$$\underbrace{7 - x_i}_{\text{first original appearances}} + \underbrace{1}_{\text{seed reinjection}} = 8 - x_i.$$

Common and markers add $g + 1 - y$, while $\sum_i x_i + y \leq 6$:

$$H \geq \sum_i (8 - x_i) + (g + 1 - y) \geq 9g - 5.$$

The grouped order A_i, B_i, C_i, D_i attains $4 + 9(g - 1) = 9g - 5$.

$H - Z^* = g$; **connected via u ; additive gap unbounded; ratio tends to $9/8$.**

Backup 2: full campaign table and provenance guards

USE FOR FAIRNESS, DENOMINATOR, MEMORY, OR VALIDATION QUESTIONS

configuration	proved / 1,410
SSPMF	1,028
CATZ-F4	876
LSS	833
BBC-LP+T	723
BBC-LP	724
BBC-K	722
BBC-LP+F+H	686
BBC-LP+F	655
BBC-LP+F+C	654
BBC-LP+ACC	652
BBC-K+F	657
BBC-LP+F+P	618

Ledger

$1,410 \times 12 = 16,920/16,920$ canonical rows; zero missing/error. Recovery replaced 123 same-envelope rows: 17 optimal, 91 time-limit feasible, 15 memory-limit. **Primary limits**

1 hour, 16 GiB. Final memory outcomes: $376 = 352$ LSS +24 BBC. Separate 32-GiB LSS sensitivity is excluded and adds zero optimum. **Fairness guards**

BBC used one CPLEX thread; compact baselines four. F5/JGSMF were not implemented. $+F$ changes presolve; $+H$ has uncharged root preprocessing. Fractional SECs were not tested.

Agreement

349,872 KTNS–DP pairs and 41,514 cross-method optimum pairs: zero disagreement.

Backup 3: the first span inequality is valid but LP-redundant

USE FOR QUESTIONS ON FUTURE FORMULATIONS

For a position window $W = [p, r]$, integer accounting gives

$$\sum_{t \in \mathcal{U}} \sum_{k=p}^r w_{t,k} \geq |\{t : a_{t,k} = 1 \text{ for some } k \in W\}| - \sum_t a_{t,p-1}.$$

But the robust fractional encoding uses $v_{t,W} \geq a_{t,k}$. Existing transition rows already imply, for every $h \in W$,

$$\sum_{k=p}^r w_{t,k} \geq \sum_{k=p}^h (a_{t,k} - a_{t,k-1}) = a_{t,h} - a_{t,p-1}.$$

Taking $\max_{h \in W}$ reproduces the added row: it cannot strengthen the LP.

Measured confirmation

- ▶ 15 windows added to PCF'.
- ▶ 37 of 37 paired roots unchanged, all at q .
- ▶ PCF' root = q on 2,108/2,108 recorded runs.
- ▶ PTF root > q on only 2/232, by one unit.

What a real strengthening must add

Tie the *union of job requirements* to a window strongly enough to survive fractional mixing, or enforce configuration-level trajectory consistency. Rewriting the same transition accounting is not enough.

A plausible integer statement is not automatically a useful LP inequality; redundancy must be proved before another large experiment.